12.6.1 Predisposing Factors for UTI

Urinary tract infection is common in females than in males. The urethra in females are shorter and wider and is less effective in preventing the bacteria entering the bladder. Sexual intercourse is a predisposing factor in females. High incidence is seen in pregnant women due to hormonal changes and impairment of urine flow due to pressure on urinary tract.



Obesity increases the risk of UTI's in men. A 2013 study examineed how obesity affected the chance of developing UTI and it was found that obese men were twice more likely to develop the UTI than obese women.

12.6.2 Urinary Tract Infection caused by *Escherichia coli*

Escherichia coli is the predominant cause of UTI.

It is a normal flora of the gut and can cause extra intestinal infections (UTI, Wound infection.) UTI (it can also be involved in other infections like wound infection peritonitis) UTI is common in (a) married women (b) elderly men with prostate enlargement.

Pathogenesis of cystitis in woman

Bladder infections can result from the downward migration of organisms from an infected kidney. But majority arise by ascent of pathogens from the rectum and vagina to the urethra meatus and bladder, leading to cystitis. If left untreated, the infection can further ascend to involve the kidneys (pyelonephritis) (Figure 12.16).

The rectum and vagina function as the reservoir of bacteria for sporadic infections

In men, the longer urethra is believed to protect against ascending infections.

When *Escherichia.coli* (and other Gram Negative rods) causes UTI, usually the number of organisms in freshly passed urine is more than 100,000 organisms/ml.

This is called "significant bacteriuria". Counts less than this is associated with contaminants from urethra or externalia. Infection of urinary tract are listed in Table 12.8.

Table 12.8: Microbial Diseases of the Urinary System

Disease	Pathogen	Symptoms
Bacterial Diseases of the Urinary system Cystitis (Urinary bladder infection)	*Escherichia* coli, *Staphylococcus saprophyticus*	Difficulty or pain in urination
Pyelonephritis (Kidney infection)	Primarily *Escherichia coli*	Fever; back or flank pain
Leptospirosis (Kidney infection)	*Leptospira interrogans*	Headaches, muscular aches, fever; kidney failure a possible complication

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